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SULTAN IDRIS EDUCATION UNIVERSITY

UNIVERSITI PENDIDIKAN SULTAN IDRIS

PENTAKSIRAN AKHIR
FINAL ASSESSMENT
SEMESTER 1 SESI 2021/2022
SEMESTER 1 SESSION 2021/2022

KOD / *CODE* : MES3103

KURSUS : PENGESAHAN DAN VERIFIKASI PERISIAN
COURSE : SOFTWARE VERIFICATION AND VALIDATION

ARAHAN / *INSTRUCTIONS*

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1. Sila baca arahan pentaksiran dengan teliti.
   Please read the assessment instructions carefully.

2. Pelajar dikehendaki menyediakan skrip jawapan/tugasan/projek mengikut kepada pentaksiran akhir yang telah ditetapkan.
   Students are required to prepare the answer scripts/assignments/projects based on the predetermined the final assessment.

3. Pelajar dikehendaki menyerahkan skrip jawapan/tugasan/projek kepada pensyarah dalam tempoh masa yang telah ditetapkan.
   Students are required to submit their answer scripts/assignments/projects to the lecturers within the specified time.

PROGRAM / *PROGRAMME*: _____

TAHUN / *YEAR*: _____ KUMPULAN KULIAH / *CLASS GROUP*: _____

NO. PENDAFTARAN / *REGISTRATION NO*: _____ | | | | | | | |

NO. KAD PENGENALAN / *I.C. NO*: | | | | | | | | | | | | |

PENSYARAH / *LECTURER*: _____

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Section A: Structured Questions (40 marks)

Instruction: Answer **all** questions in the space provided.

1. Ali is a former IT Executive in one of the Nuclear Agencies in Australia. Now, he worked as a Chief Technology Officer (CTO) at SBA EVOLUTION. Sdn. Bhd in Malaysia. This company is a developer for educational digital teaching aids. Ali noticed that this company rarely conducts verification and validation (V&V) for their software development. However, this company do not void any practice because level of effort of organization to expend on V&V should be commensurate with the size and criticality of software. By using Socratic approach, four (4) questions that you should ask yourself to assess how much verification and validation is required. State **three (3)** of them?

[3 marks]

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2. In your own words, justify with example, why verification and validation (V&V) process is important?

[2 marks]

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3. Explain **five (5)** differences between verification and validation (V&V) according to the Table 1.

Table 1: Differences between verification and validation

Description	Verification	Validation
Definition		
Evaluation period		
Objective		
Activities included/Methods		
Items evaluated/Target		

[5 marks]

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4. Ahmad was appointed as a project manager to measure the quality of the HealthCare system that developed by Pusat Kesihatan UPSI for detecting the COVID-19 diseases. He should provide a report to ensure the system is reliable to be delivered. Ahmad plans to use McCalls's factor model to measure software quality. The report must include all factors that belong to the Model. As an assistant to Ahmad, he ask you to answer this question: 1) What are **three (3)** factor categories of McCalls's factor model?; and 2) Elaborate **two (2)** factors for each category.

[10 marks]

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5. Choose **one (1)** factor that best fit each requirement below according to McCalls's factors and to which categories does the factor belongs. (Please state the answer in Table 2)

Table 2: Requirements based on McCall's Factor Model

Requirements	**McCall's Requirement Factor Category**	**Factor**
The probability a software system will be found in a state of failure during peak hours (9am-4pm) is required to be below 0.5%.		
The software package developed for the Linux operating system should be compatible for applications in a Windows NT environment.		

[5 marks]

Q6. Briefly describe **five (5)** various causes of software errors.

[5 marks]

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Q7. Black box and White Box are two (2) methods for software testing in validating the actual product. Explain **five (5)** differences between black box testing and white box testing.
(No long explanation is required so you may use a table to represent your answer)

[10 marks]

END OF QUESTIONS

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